E-Commerce Sales Drop Analysis

From revenue-drop brief to validated trend, driver analysis, Power BI dashboard and interview story

A practical project execution log • Synthetic portfolio case study

1. Executive summary

The business question is: why did e-commerce revenue fall in the latest months? On rows with a valid order date, revenue was ■9,587,109 in July 2026, ■7,046,323 in August, and ■6,169,742 in September. July to September revenue changed by -35.6%; orders changed from 2,600.0 to 1,768.0 (-32.0%). The pattern is a real decline in the dated subset, but 800 rows with ■2,687,509 revenue have unusable dates and cannot be assigned to a month until resolved.

Treat these as descriptive findings from synthetic data, not real company performance. The main diagnosis is that both order volume and revenue fell; the next step is to decompose by channel, product, region, discount, and customer segment before claiming a root cause.

Metric	Jul 2026	Aug 2026	Sep 2026	Jul → Sep
Revenue	■9,587,109	■7,046,323	■6,169,742	-35.6%
Orders	2,600.0	2,069.0	1,768.0	-32.0%
Quantity	3,395.0	2,541.0	2,180.0	-35.8%
Estimated profit	■2,686,645	■1,989,075	■1,721,840	-35.9%

2. Project brief and starting point

Business context

A hypothetical e-commerce team reports that revenue has declined recently. The job is not to immediately blame marketing or a product; it is to define the time window, verify the metric, find where the decline is concentrated, and recommend tests supported by evidence.

Question tree

• Is the decline real, or is it caused by missing dates, status definitions, or duplicate orders?

• Is revenue down because fewer orders are placed, average order value falls, or both?

• Which channel, category, region, and customer segment explain the largest percentage and absolute changes?

• Did discounting, product mix, or estimated margin change at the same time?

Metric definitions

Metric	Definition	Why it matters
Revenue	Sum of order revenue for valid dated rows	Primary business outcome
Orders	Distinct order_id count	Demand / transaction volume
AOV	Revenue ÷ distinct orders	Separates volume from basket value
Quantity	Sum of quantity	Units sold; helps explain product volume
Estimated profit	Sum of profit_estimate	Revenue decline may not equal profit decline

3. Chronological execution log: what I did and why

Step	Action	Reason / decision rule	Output
1. Frame	Turned “revenue fell” into measurable question; selected July–September as recent window	Avoid comparing unrelated months or cherry-picking a peak	Defined comparison window

Step	Action	Reason / decision rule	Output
2. Inspect	Checked row counts, columns, dates, IDs, status, missingness, duplicates	Validate whether dataset can answer the question	Quality checklist
3. Validate	Reconciled monthly revenue and distinct order counts	Catch broken formulas, missing dates, or duplicate IDs	Monthly trend
4. Clean	Parsed order_date; created month from parsed date; kept undated rows separate	Do not silently invent dates or put them in arbitrary months	Valid-date analysis subset
5. Enrich	Joined orders to customer and product lookup tables by IDs	Enable region, segment, category analysis	Joined analytical table
6. Decompose	Compared revenue/orders/AOV by month and dimensions	Find contribution and rate changes, not just biggest group	Driver analysis
7. Visualise	Built KPI cards, monthly trend, breakdown charts, matrix	Make decline and drivers easy to inspect	Power BI report
8. Recommend	Matched action to verified driver and defined test metric	Avoid recommendations not supported by evidence	Action plan

Data audit result: 800 of 32,000 rows have invalid/missing order dates; these rows contain ■2,687,509 revenue. 128 order rows have customer IDs not found in the cleaned customer table. Do not hide either issue. Keep undated revenue as a separate reconciliation KPI and report customer join coverage.

Cleaning decisions: standardise dates and categories, preserve source IDs, remove only confirmed duplicate transaction records (not repeated product lines without checking the grain), validate quantity and prices as numeric, recalculate revenue as quantity × unit price × (1 – discount_pct) where that is the agreed business rule, and compare calculated versus supplied revenue before replacing anything. In this supplied file, the final cleaning is not fully production-grade because the undated rows and unmatched customer IDs remain; document them as unresolved data-quality exceptions.

4. SQL investigation (with explanations)

Assume tables are orders_clean, customers_clean, and products_clean. SQL syntax is broadly SQLite-compatible; date functions may differ by database.

A. Monthly trend

```
SELECT substr(order_date, 1, 7) AS month,
COUNT(DISTINCT order_id) AS orders,
SUM(revenue) AS revenue,
SUM(revenue) * 1.0 / COUNT(DISTINCT order_id) AS aov
FROM orders_clean
WHERE order_date IS NOT NULL
GROUP BY substr(order_date, 1, 7)
ORDER BY month;
```

Why: aggregate to month, count distinct transactions, and compute AOV so revenue movement can be separated into volume and basket size.

B. Recent-month comparison by channel

```
SELECT substr(order_date,1,7) AS month, sales_channel,
COUNT(DISTINCT order_id) AS orders, SUM(revenue) AS revenue
FROM orders_clean
WHERE order_date >= '2026-07-01' AND order_date < '2026-10-01'
GROUP BY substr(order_date,1,7), sales_channel
ORDER BY month, revenue DESC;
```

Why: a channel's absolute revenue may be large simply because it is large overall. Compare month-over-month percentage changes and contribution to total decline, not just totals.

C. Check unmatched customer keys

```
SELECT COUNT(*) AS unmatched_order_rows
FROM orders_clean o
LEFT JOIN customers_clean c ON o.customer_id = c.customer_id
WHERE c.customer_id IS NULL;
```

Why: a left join preserves orders and reveals missing lookup coverage. Never use an inner join without checking whether it drops material revenue.

5. Excel / spreadsheet workflow

• Import the cleaned CSVs; confirm dates are dates and revenue/discount are numeric.

• Create a pivot table: Rows = month; Values = Sum of revenue, distinct count of order_id if available, sum of quantity.

• Add AOV = revenue / orders; calculate month-over-month % change = (current month – prior month) / prior month.

• Create separate pivots by sales_channel, product category, region, customer_segment.

• Use a reconciliation tab listing invalid dates, unmatched customer IDs, duplicates, and revenue affected.

Expected output: a traceable workbook where each chart total ties back to source rows and the exceptions are visible.

6. Power BI model and dashboard

Model

Use a star-style model: Orders as the fact table; Customers and Products as dimensions. Use one-to-many relationships from Customers[customer_id] to Orders[customer_id] and Products[product_id] to Orders[product_id]. Add a proper Calendar table and relate it to Orders[order_date]. Keep invalid dates out of the calendar relationship and expose them separately.

Starter DAX

```
Revenue = SUM(Orders[revenue])
Orders = DISTINCTCOUNT(Orders[order_id])
Units Sold = SUM(Orders[quantity])
AOV = DIVIDE([Revenue], [Orders])
Estimated Profit = SUM(Orders[profit_estimate])
MoM Revenue % =
VAR PrevMonth = CALCULATE([Revenue], DATEADD('Calendar'[Date], -1, MONTH))
RETURN DIVIDE([Revenue] - PrevMonth, PrevMonth)
Undated Revenue = CALCULATE([Revenue], ISBLANK(Orders[order_date]))
```

Dashboard pages

Page	Visuals	Question answered
Executive overview	Revenue, orders, AOV, estimated profit cards; monthly line chart	When did the decline start and which KPI moved?
Drivers	Revenue and orders by channel/category/region; matrix with % change	Where is the decline concentrated?
Customer & product	Segment, acquisition channel, product category, top movers	Which customer/product groups shifted?
Data quality	Missing dates, unmatched IDs, duplicate checks	How much can we trust the conclusions?

Add slicers for month, channel, region, category, and segment. Use consistent date range and display both absolute change and percentage change. Avoid showing a monthly trend that silently excludes invalid-date revenue without saying so.

7. Findings, interpretation, and recommendations

Verified findings

• On valid-date rows, revenue fell 35.6% from July to September 2026; orders fell 32.0% and units fell 35.8%. This points to a transaction/volume decline, not only a price effect.

• From August to September, revenue changed -12.4% and orders changed -14.5%, so the decline continued into September.

• ■2,687,509 of revenue is tied to 800 rows without valid dates; the total-period revenue picture is incomplete until these are resolved.

Do not claim a channel or product caused the decline until you have compared channel/category contribution to the month-to-month change. The prepared data supports this next step, but the root cause is not proven by the headline totals alone.

Recommendations

• First, reconcile the 800 undated orders and unmatched customer IDs. This is a decision-quality issue, not housekeeping.

• Decompose the July-to-September revenue gap by channel, category, region, and customer segment; report absolute gap and % change side by side.

• If order volume is the main driver, test targeted acquisition/reactivation or merchandising changes in the affected segment rather than blanket discounts.

• Track weekly orders, conversion, AOV, cancellation/refund rate (if available), and contribution margin. Use the same baseline and compare with a prior period of equal length.

Limitations

• Synthetic dataset; no external seasonality or marketing-spend data; profit is estimated; date quality issues remain; correlation by segment does not establish causation.

8. Interview preparation

60-second project story

“I investigated a recent e-commerce revenue decline using synthetic order, customer, and product data. I first defined a recent comparison window and audited the data, then built monthly revenue, order, quantity, and AOV views. The valid-date subset showed revenue and order volume both falling from July to September. I also found 800 undated order rows containing material revenue and unmatched customer keys, so I kept those as explicit limitations instead of forcing them into the trend. My next diagnostic step was a contribution analysis by channel, category, region, and segment; I would only recommend a specific growth intervention after the affected dimension is verified.”

Questions you should be able to answer

• Why use both absolute and percentage change? Absolute change shows business impact; percentage change allows fair comparison across differently sized groups.

• Why not compare September with October 2025? That may mix separate trend phases and seasonal effects; a recent equal-length baseline better matches the ticket, while year-over-year analysis would be a separate check.

• What is AOV? Revenue divided by distinct orders; it helps separate fewer orders from smaller baskets.

• Why keep missing dates visible? Assigning a made-up date could create a false trend. Reconcile source systems or exclude from month-level analysis and disclose the amount.

• What would you do next? Validate date exceptions, run dimensional contribution analysis, inspect weekly data, and test hypotheses with operational/marketing evidence.